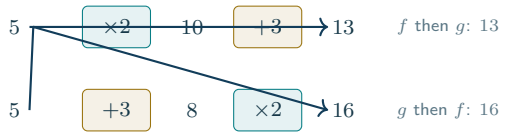
ALGEBRA LADDER 5 OF 9

Actions that do not commute: composition

Doing one thing and then another is an operation, and it keeps associativity while giving up commutativity.

1 / THE RULE AND THE PICTURE

Two machines in the opposite orders can give different answers



Let f double and g add 3. Writing $g \circ f$ for "f first, then g", the same starting number 5 ends at 13 one way and at 16 the other:

$$\boxed{g \circ f \neq f \circ g}$$

because $g(f(x)) = 2x + 3$ and $f(g(x)) = 2(x + 3) = 2x + 6$.

2 / WHICH LAWS SURVIVE

Order matters, grouping does not, and every reversible action can be undone

law for composition	verdict and example
associative	yes: $(h \circ g) \circ f = h \circ (g \circ f)$, both equal $(2x + 3)^2$ in the example
commutative	no: $g \circ f = 2x + 3$ but $f \circ g = 2x + 6$
identity	yes: the function that returns its input, $\mathrm{id}(x) = x$
inverse	yes for a function that can be reversed, and then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$
distributes over $+$	only for functions like $f(x) = 2x$: $f(1 + 2) = f(1) + f(2) = 6$ but $g(1 + 2) = 6$ and $g(1) + g(2) = 9$

3 / WALKING AND TURNING

The same pair of moves in the other order ends somewhere else

Starting at the origin facing north, "turn right and then walk 10 steps" ends at the point 10 east and 0 north, while "walk 10 steps and then turn right" ends 10 north and 0 east, so the two orders land at different places.
The sock and shoe example is the same idea, since the second action only makes sense after the first.

4 / UNDOING A PAIR OF ACTIONS

The undo reverses the order

To undo "socks then shoes" we take the shoes off first and then the socks, and for the machines, $g \circ f$ turns x into $2x + 3$, so its inverse takes y to $\frac{y-3}{2}$, which is $f^{-1}(g^{-1}(y))$ with the subtraction first and the halving second. Reversing the order gives $g^{-1}(f^{-1}(y)) = \frac{y}{2} - 3$, a different answer, and the check $y = 13 \to 5$ works only for the first.

5 / TRY THESE

Compose in both orders and compare

(a) For $f(x) = x + 1$ and $g(x) = x^2$, find $g \circ f = (x + 1)^2$ and $f \circ g = x^2 + 1$, which differ at $x = 2$ as 9 and 5.
(b) Is "add 3 then add 5" the same as "add 5 then add 3"? Yes, because adding is commutative, so some pairs of actions do commute.
(c) Find a pair of machines that commute: $f(x) = 2x$ and $g(x) = 5x$ give $10x$ in either order.

ASIDE / A NEW KIND OF PRODUCT

Composition is a product of actions with an identity, inverses and associativity, so it behaves like numbers except for commutativity.

LINKS TO OTHER SHEETS

The next sheet lists every way of turning and flipping a triangle and writes the products in a table; the matrix sheet shows composition as multiplication.

Convention. $g \circ f$ means apply f first and then g; id is the identity function; f^{-1} undoes f.